

Probability and Conditional Probability

Example 1

You have two books. You have a probability of 0.3 of liking book A, and a probability of 0.4 of liking book B. You have a probability of 0.2 of liking both book A and B. What is the probability that you like neither book?

Example 2

The game of 'Toto' is a football match outcome guessing game played in some European countries. There are 13 football matches and hence 13 entries. In each entry, you can tick it is a 'home win', 'draw' or 'away win'.

The prize money depends on how many of your guesses are correct.

If you are just guessing,

- a) what is the probability that all your 13 guesses are correct?
- b) what is the probability that 11 of your guesses are correct?

Example 3

Probability is readily applied in games of chances.

Poker is played with a 52 card deck (13 cards of 4 suits). A player's hand is composed of 5 cards.

A 'royal flush' is the most difficult hand to obtain. It consists of Ace, King, Queen, Jack and 10, all of the same suit.

A 'pair' is the hand of the lowest value. It contains a simple pair of cards of the same value. An example is Ace Ace Four Jack Nine.

When the cards are dealt at the beginning of the game,

- a) what is the probability of obtaining a "royal flush"?
- b) what is the probability of obtaining a "pair"?

Example 4

Mark 6 consists of 49 numbers. 6 numbers are randomly drawn without replacement. Then an extra number is drawn without replacement.

4th price is won when you have guessed right on 4 numbers as well as the extra number.

Calculate the probability of winning the 4th price.

Notes about this question:

In Mark Six, people fill in 6 numbers only in the entry form. For the fourth price, 4 out of 6 numbers are correct, the 5th number is incorrect, and the 6th number must be the extra number.

https://en.wikipedia.org/wiki/Mark_Six

Prize	Criteria
1st	All 6 drawn numbers
2nd	5 out of 6 drawn numbers, plus the extra number
3rd	5 out of 6 drawn numbers
4th	4 out of 6 drawn numbers, plus the extra number
5th	4 out of 6 drawn numbers
6th	3 out of 6 drawn numbers, plus the extra number
7th	3 out of 6 drawn numbers

Example 5

Assume 45% of CityU undergraduates are male and 30% of CityU undergraduates are in the College of Engineering (CENG). Assume 70% of undergraduate students in CENG are male. If a student is selected at random, find the conditional probability that

- a CityU undergraduate student is male, given that the student is a CENG student.
- the student is in CENG, given that the student is male.

Example 6

- a) Find the probability of a plane crash.
- b) If a plane crash occurs, find the probability of survival.
- c) Hence find the probability of dying in a plane crash.

Counter-intuitive Results (Apparent paradox) in Probability

Example 7

The Monte Hall Problem: <https://www.youtube.com/watch?v=mhlc7peGlGg>. Stop when the host gives you 10 seconds. Think about it carefully before continuing.

Suggest a way to make sense of the counter-intuitive result of the Monte Hall problem.

Example 8

Assume that the probability of having a boy and a girl is equal and the events are independent.

- a) What is the probability that a family with two children has at least one boy?
- b) Suppose we know that the family has a boy, what is the probability that the other child is also a boy?
- c) Suppose we visit a family with two children. We knock the door. Assume that it is equally probable that the first or the second child show up. Suppose a boy shows up. What is the probability that the other child is also a boy?

Example 9

Assume for simplicity that a year has 365 days, i.e., we can ignore leap year. Suppose the probability that a person is born on any of the 365 days is equal and independent. There is a party of n persons. How large should n be such that the probability that two persons in the party have the same birthday is larger than 50%?

Bayes' Rule

Example 10

In the example concerning laboratory blood test in the lecture notes, if the disease is very rare and 1 in 1 million (1×10^{-6}) chance of the population has the disease, what is the probability that a person has the disease given that the test result is positive?

Look up https://en.wikipedia.org/wiki/Rare_disease for the definition of rare disease.

Example 11

The following web page gives the statistics of cancer in Hong Kong:

<https://www3.ha.org.hk/cancereg/topten.html>

A patient comes to see a doctor and describes his symptoms. Based on the doctor's experience, he is 60% certain that the patient may have cancer. He asks the patient to conduct a test, which will give a cancer index number. A number that exceeds the threshold indicates cancer and vice versa.

Suppose that the test has a false positive rate of 10% and a false negative rate of 1%. What is the conditional probability that the patient has the cancer given that

- a) the test indicates that he has the cancer?
- b) the test indicates that he does not have the cancer?
- c) Do you think the test is useful? Comment.

Example 12

At a certain stage of criminal investigation, the inspector in charge is 60% convinced of the guilt of a certain suspect. Suppose now that a new piece of evidence that shows the criminal has a certain characteristic is uncovered. If 20% of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has this characteristic?

Example 13

Let us now suppose that the new evidence is subject to different possible interpretations, and in fact only shows that it is 90% likely that the criminal possesses this certain characteristic. In this case, how likely would it be that the suspect is guilty (assuming, as before, that he has this characteristic)?

Example 14

A plane is missing and it is presumed that it was equally likely to have gone down in any of three possible regions. Let $1 - \alpha_i$ denote the probability that the plane will be found upon a search of the i th region when the plane is, in fact, in that region, $i = 1, 2, 3$. (The constant α_i are called overlook probabilities because they present the probability of overlooking the plane; they are generally attributable to the geographical and environmental conditions of the regions.) What is the conditional probability that the plane is in the i th region, given that a search of region 1 is unsuccessful, $i = 1, 2, 3$?

Example 15

A binary classifier is one which classifies an input instance into “positive” or “negative”. The quality of a binary classifier can be expressed by its

- a) accuracy
- b) precision
- c) recall
- d) specificity

Find the definitions of these terms. What are the meanings of these terms?

Detecting Independent Events

Example 16

Two fair die are thrown. Let E be the event that the second throw is 6 and let F be the event that the sum of the two throws is 10. Prove or disprove that the two events are independent.

Example 17

Two fair dice are thrown. Let E be the event that the sum is 8. Let F be the event that the first throw is 2. Show that the two events are not independent.